



PARKINSON'S MOTOR DIARY

A diary that tells you *when it doesn't know.*

Reconstructs the hours a Parkinson's symptom diary never recorded, from two wrist sensors — and refuses, hour by hour, wherever the evidence cannot carry an answer.

github.com/MohammadSadeghSalehi/astrolabe

THE PROBLEM

Treatment depends on knowing *when*. Nobody records it.

Symptoms fluctuate hour to hour. The clinic visit that sets medication sees one of those hours. The diary meant to fill the gap is a chore, so it stops getting filled.

11.8M

people living with Parkinson's

up 274% since 1990 · *Lancet Healthy Longevity*
2024

60–90%

develop motor fluctuations

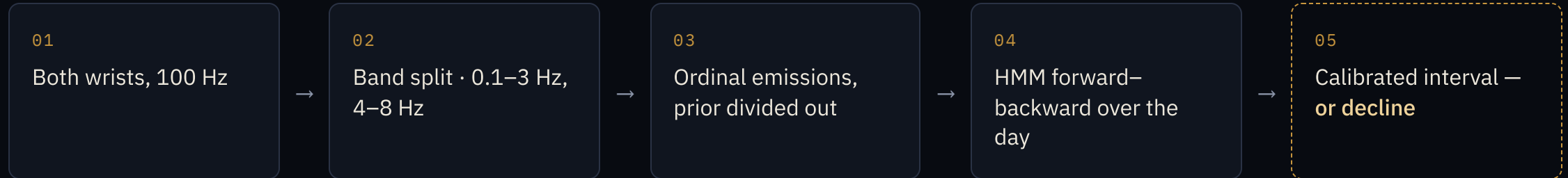
within 5–10 years of levodopa · *PMC4634347*

61.6%

of hours carry a diary entry

and that is *with* research staff helping

One signal, five stages, and a refusal.

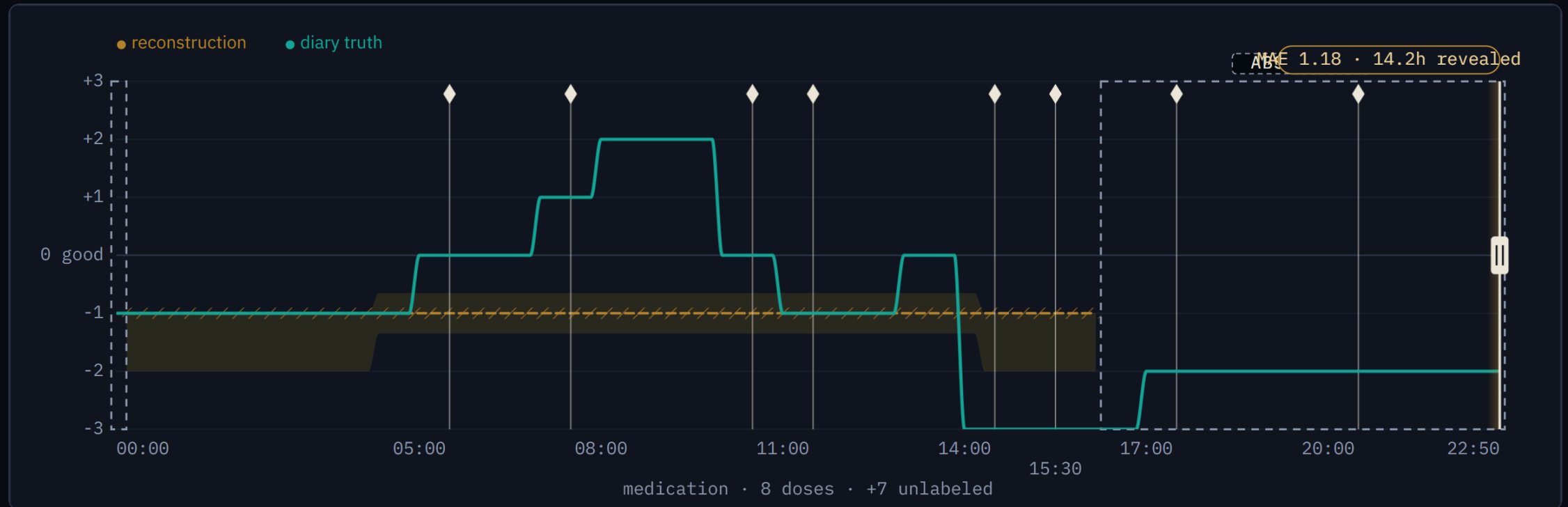


Coverage is earned, not asserted. Asking for 90% of posterior mass does not give 90% empirical coverage. The interval is widened on held-out people until measured coverage reaches target.

Abstention holds an error budget, not a rate. Tune to a fixed refusal rate and a degraded sensor *cannot* show degradation. Fix the error budget and a worse one can only comply by answering less.

THE DEMO

Drag, and the real diary comes out from behind the reconstruction.



A participant **held out of training** – the model is meeting them for the first time.

On this day it answers nothing — and says why.



114 of 114 steps declined. Peak confidence 0.55; it answers at 0.58. That is the system working.

MEASURED ON PEOPLE THE MODEL NEVER TRAINED ON

What holds up — and what does not.

HOLDS

The 90% interval really contains the truth	0.903	against a 0.90 target
Refusing improves what remains	0.713 → 0.825	all hours vs the most confident quarter
Losing a wrist costs answers, not accuracy	12.4% → 77.3%	hours declined, same error budget

DOES NOT

7-state motor scale, ordinal MAE	0.684	worse than a 0.594 constant
Median <i>within-participant</i> tremor AUC	0.550	the figure a diary is judged on

So we do not predict how the day felt. That row stays yours — what you reported, or an honest blank. It is on our landing page, not in a footnote.



HOW IT WAS BUILT

Three agents, one frozen contract, no model near a prediction.

- ML, UI and the film behind a frozen JSON contract, so no agent could reshape another's output
- **ElevenLabs** transcribes the voice note, so a tremor does not mean typing
- **OpenAI** structures it — behind a guard rejecting any number not in the transcript
- Hand-written maths, hand-rolled SVG. Every figure regenerates from a named script

github.com/MohammadSadeghSalehi/astrolabe

runs locally — no keys, no database

Not a medical device. No diagnostic, dosing or treatment claim is made. COPS data CC-BY 4.0
— 66 people with Parkinson's, bilateral wrist accelerometry.

ASTROLABE